

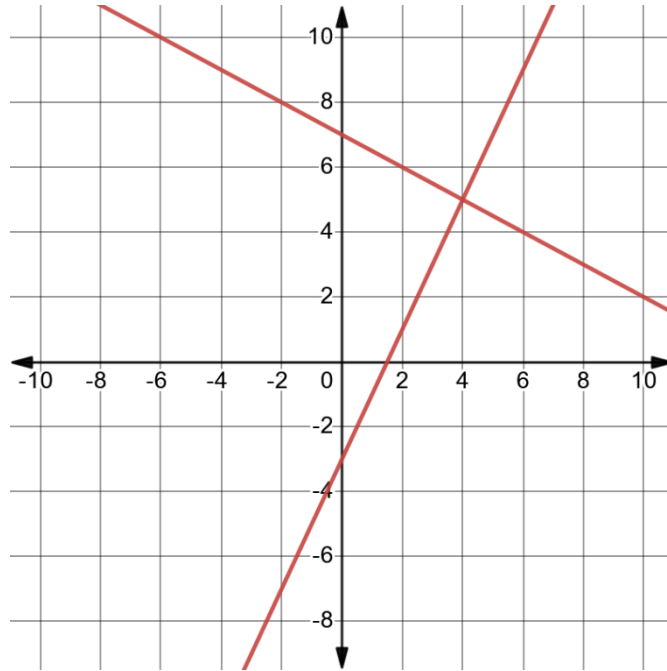
Geometry
Unit 13
Lesson 1 Practice

Name ___ **Answer Key** ___
Date _____
Period _____

1. Graph the equations on the coordinate plane:

$$y = -\frac{1}{2}x + 7$$

$$y = 2x - 3$$



2. The equations $y = -\frac{1}{2}x + 7$ and $y = 2x - 3$ resulted in

☐ parallel
☒ **perpendicular**
☐ vertical

lines.

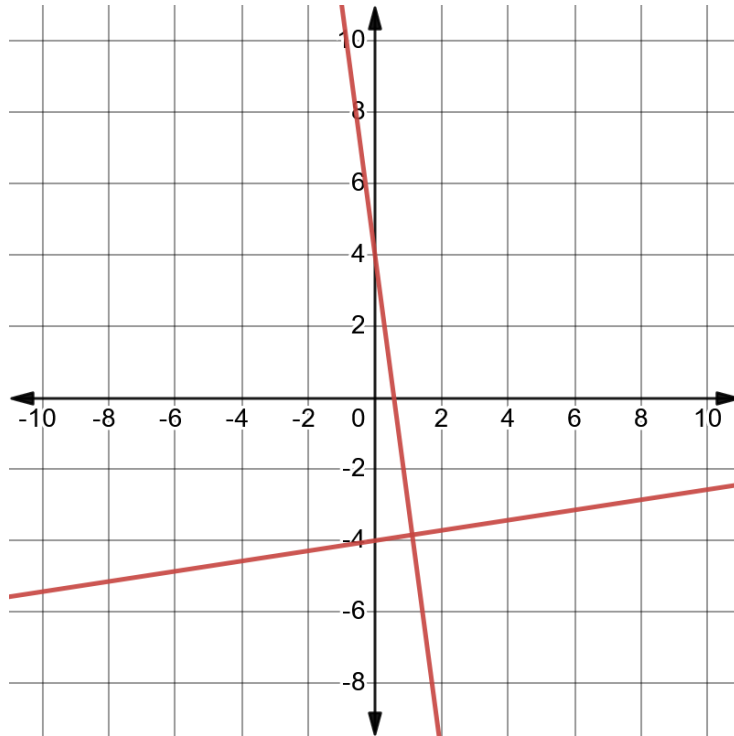
3. The slopes of the equations are

☐ equal.
☐ undefined.
☒ **opposite reciprocals.**

4. Rewrite the equation $7y - x = -28$ in slope intercept form and graph the equation using slope-intercept form.

$$7y = x - 28$$

$$y = \frac{1}{7}x - 4$$



5. Determine if $y = -7x + 4$ is parallel, perpendicular, or neither to the graph of the equation $7y - x = -28$.
- $$y = -7x + 4 \rightarrow m = -7$$
- $$7y - x = -28 \rightarrow m = \frac{1}{7}, \text{ so perpendicular}$$

6. Graph the equation $y = -7x + 4$ on the graph to verify your answer to question 5.
See graph.

7. Which equation represents a line which is perpendicular to the line $y = \frac{3}{8}x - 1$?

$$m = -\frac{8}{3}$$

☐ $3x + y = -40$

☐ $3x - 8y = -56$

☐ $3y - 8x = 3$

☒ $8x + 3y = 9$

$$-8y = -56 - 3x \rightarrow m = \frac{3}{8}$$

$$3y = 3 + 8x \rightarrow m = \frac{8}{3}$$

$$3y = 9 - 8x \rightarrow y = 3 - \frac{8}{3}x \rightarrow m = -\frac{8}{3}$$

8. Which equation represents a line which is parallel to the line $y = -\frac{2}{3}x + 1$?

$$m = -\frac{2}{3}$$

☐ $2x - 3y = 21$

☐ $2y - 3x = 2$

☐ $3x - 2y = -2$

☒ $2x + 3y = -21$

$$-3y = 21 - 2x \rightarrow y = -7 + \frac{2}{3}x \rightarrow m = \frac{2}{3}$$

$$2y = 2 + 3x \rightarrow m = \frac{3}{2}$$

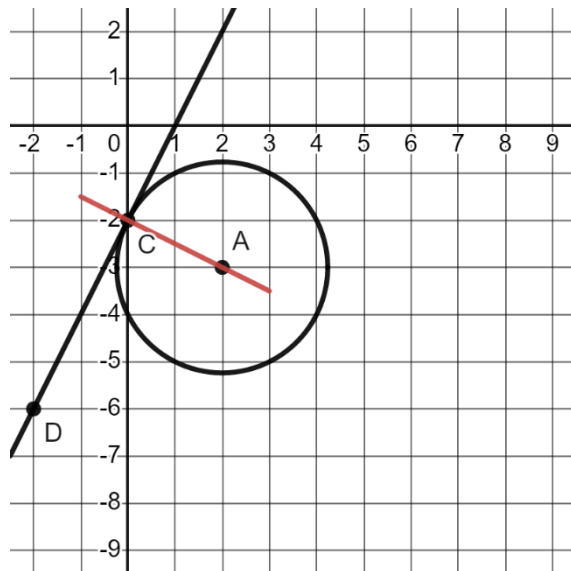
$$-2y = -2 - 3x \rightarrow m = \frac{3}{2}$$

$$3y = -21 - 2x \rightarrow y = -7 - \frac{2}{3}x \rightarrow m = -\frac{2}{3}$$

Geometry
Unit 13
Lesson 2 Practice

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Use the following information for questions 1-4. Circle A is shown, where point C is on the circle and line CD is tangent to circle A .



1. Use a straightedge to draw a line that goes through points A and C .

2. Write the equation of line CD .

Slope: 2

$$y - y_1 = m(x - x_1)$$

$$y - (-6) = 2(x - (-2))$$

$$y + 6 = 2(x + 2)$$

$$y = 2x - 2$$

3. Write the equation of line AC .

Slope: $-\frac{1}{2}$

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = -\frac{1}{2}(x - 0)$$

$$y + 2 = -\frac{1}{2}x$$

$$y = -\frac{1}{2}x - 2$$

4. Line CD and line AC intersect at point $(0, -2)$. Use the equations of line CD and AC to confirm this point.

$$y = 2x - 2$$

$$-2 = 2(0) - 2$$

$$-2 = -2 \checkmark$$

$$y = -\frac{1}{2}x - 2$$

$$-2 = -\frac{1}{2}(0) - 2$$

$$-2 = -2 \checkmark$$

Use the following information for questions 5-8. Circle P has a center at $(4, -3)$ and a tangent line with the equation $y = 4x - 2$.

5. Write the equation of the line perpendicular to the tangent line.

Slope: $\frac{4}{1} \rightarrow \frac{1}{4} \rightarrow -\frac{1}{4}$

$y = -\frac{1}{4}x - 2$

6. Find the point where the tangent line touches circle P . Call the point H .

H is $(0, -2)$

7. What is the radius of circle P ?

$r = \sqrt{17}$

8. Write the equation of circle P .

$(x - 4)^2 + (y + 3)^2 = 17$

Use the following information for questions 9-10. Circle A is shown with tangent lines PK and DK .

9. Complete the table.

Line Segment	Length
$\overline{PK}$	$\sqrt{325}$
$\overline{DK}$	$\sqrt{325}$
$\overline{AD}$	$\sqrt{13}$

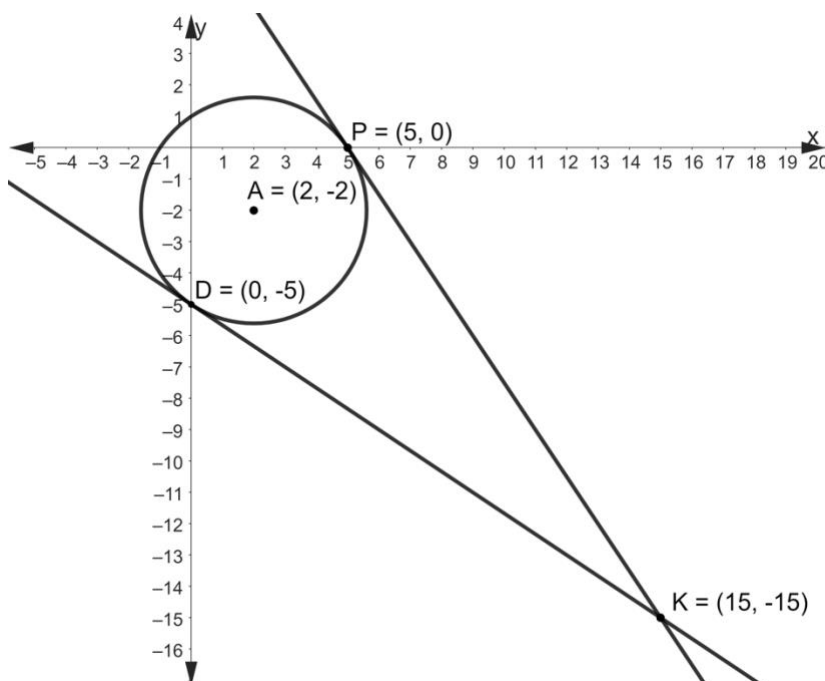
$10^2 + 15^2 = (PK)^2$

$10^2 + 15^2 = (DK)^2$

$(-3)^2 + 2^2 = (AD)^2$

10. Write the equation of circle A .

$(x - 2)^2 + (y + 2)^2 = 13$



Geometry
Unit 13
Lesson 3 Practice

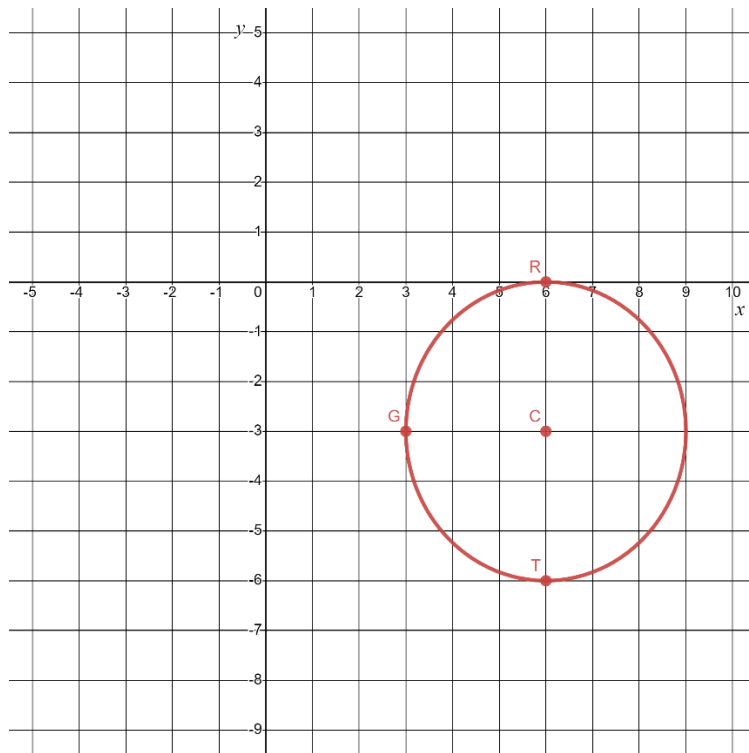
Name Answer Key
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The equation of circle C is $(x - 6)^2 + (y + 3)^2 = 9$.

1. Complete the table by substituting the given coordinates for x and y in the equation of circle C . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$R(6,0)$	$(6 - 6)^2 + (0 + 3)^2 = 9$ $(0)^2 + (3)^2 = 9$	$9 = 9$
$G(3, -3)$	$(3 - 6)^2 + (-3 + 3)^2 = 9$ $(-3)^2 + (0)^2 = 9$	$9 = 9$
$T(6, -6)$	$(6 - 6)^2 + (-6 + 3)^2 = 9$ $(0)^2 + (-3)^2 = 9$	$9 = 9$

2. Graph circle C , $(x - 6)^2 + (y + 3)^2 = 9$.



3. Plot $R(6,0)$, $G(3,-3)$, $T(6,-6)$ on the circle.
4. Determine the length of the diameter on circle C .
Diameter = 6

Circle D has a radius of 2 with points $L(-3,7)$, $F(-1,5)$, and $T(1,7)$ on the circle.

5. Determine which two points are the endpoints of a diameter of circle D .
 L and T

6. Find the center of circle D .
 $(-1,7)$

7. Write the equation for circle D .
 $(x + 1)^2 + (y - 7)^2 = 4$

8. The equation of circle W is $(x + 7)^2 + y^2 = 4$. Determine whether the following points lie inside, outside, or on circle W .

Point	Work	Location
$C(-7,2)$	$(-7 + 7)^2 + (2)^2 = 4$ $(0)^2 + 4 = 4$ $4 = 4$	☐ Inside circle W ☐ Outside circle W ☒ On circle W
$N(-6,1)$	$(-6 + 7)^2 + (1)^2 = 4$ $1 + 1 = 4$ $2 = 4$	☒ Inside circle W ☐ Outside circle W ☐ On circle W
$T(-7,5)$	$(-7 + 7)^2 + (5)^2 = 4$ $(0)^2 + 25 = 4$ $25 = 4$	☐ Inside circle W ☒ Outside circle W ☐ On circle W

9. The equation of circle R is $(x - 4)^2 + (y + 2)^2 = 13$.

Select all points that lie on circle R .

- ☒ $(6,1)$ $(6 - 4)^2 + (1 + 2)^2 = 4 + 9 = 13$
☐ $(4,-2)$
☒ $(1,0)$ $(1 - 4)^2 + (0 + 2)^2 = 9 + 4 = 13$
☐ $(0,0)$
☐ $(-4,-2)$

10. Circle M and circle T intersect at point H .

Equation of circle M is $x^2 + (y - 2)^2 = 10$

$$(-3)^2 + (1 - 2)^2 = 9 + 1 = 10 \checkmark$$

Equation of circle T is $(x + 4)^2 + (y - 2)^2 = 2$

$$(-3 + 4)^2 + (1 - 2)^2 = 1 + 1 = 2 \checkmark$$

Which coordinate pair is the location of point H .

- ☐ $(-5,2)$
☐ $(0,0)$
☐ $(-3,0)$
☒ $(-3,1)$

Geometry
Unit 13
Lesson 4 Practice

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Use the following information for questions 1-4. Directed line segment $\overline{LT}$ is shown.



- How many equal parts is $\overline{LT}$ partitioned into? **6**
- Make a mark where $\overline{LT}$ is partitioned into a 1:3 ratio. Call this point K .
- Complete the statement.

Point K is $\frac{1}{4}$ of the distance from L to T .

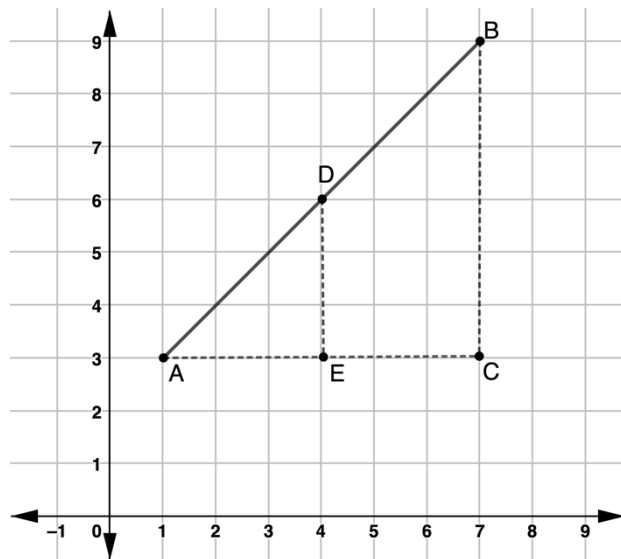
- Complete the equation that can be used to find the distance of $\overline{LK}$.

$$LK = \frac{1}{4} LT$$

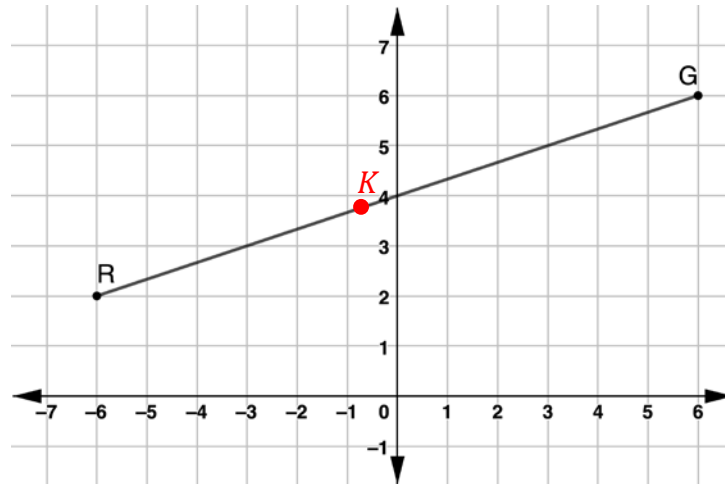
- Point D partitions directed line segment $\overline{AB}$ in a 3:3 ratio. Points $A(1,3)$, $E(4,3)$, $D(4,6)$, $B(7,9)$, and $C(7,3)$ are shown on the coordinate grid.

Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represent.

Segment Ratio	Numeric Ratio	Type of Ratio
$AE:EC$	3:3	☐ part: part ☐ part: whole
$AE:AC$	3:6	☐ part: part ☐ part: whole
$DE:BC$	3:6	☐ part: part ☐ part: whole
$AD:DB$	3:3	☐ part: part ☐ part: whole



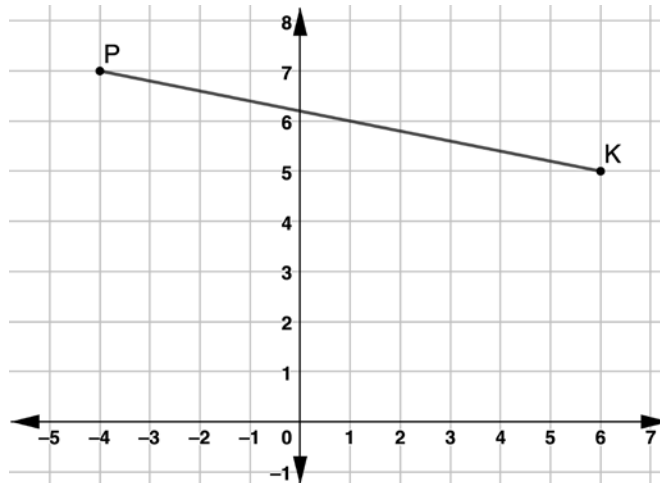
Use the following information for questions 6-8. Directed line segment RG is shown with endpoints at point $R(-6,2)$ and point $G(6,6)$.



Directed line segment RG is partitioned $\frac{3}{4}$ of the distance from R to G at point T .

6. What is the horizontal distance of $\overline{RG}$? $3\left(\frac{12}{7}\right) = \frac{36}{7}$
7. What is the vertical distance of $\overline{RG}$? $3\left(\frac{4}{7}\right) = \frac{12}{7}$
8. What are the coordinates of point T ? $\left(-\frac{6}{7}, \frac{26}{7}\right)$

Use the following information for questions 9-10. Directed line segment PK has endpoints at $P(-4,7)$ and $K(6,5)$.



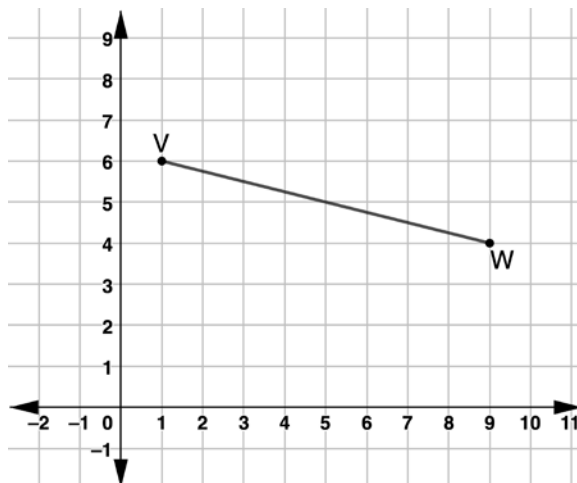
Directed line segment PK is partitioned $\frac{2}{5}$ of the distance from P to K at point R .

9. What is the horizontal and vertical distance of $\overline{PK}$? $\frac{20}{7}$ $-\frac{4}{7}$
10. What are the coordinates of point R ? $\left(-\frac{8}{7}, \frac{45}{7}\right)$

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Lesson 5 Practice

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Use the following information for questions 1-4. Point V has coordinates (1,6) and point W has coordinates (9,4).



1. Complete the table to calculate $\frac{1}{2}V + \frac{1}{2}W$.

	x-value	y-value
$W(1,6)$	$\frac{1}{2}(1)$	$\frac{1}{2}(6)$
$V(9,4)$	$\frac{1}{2}(9)$	$\frac{1}{2}(4)$
Total	5	5

2. What is the midpoint of line segment VW ?

$$\text{Midpoint} = \left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2} \right) = \left(\frac{9+1}{2}, \frac{6+4}{2} \right) = \left(\frac{10}{2}, \frac{10}{2} \right) = (5,5)$$

3. Complete the expression that can be used to find the point that partitions the line segment in a 1:1 ratio.

$$\frac{1}{2}V + \frac{1}{2}W$$

4. Find the point that partitions directed line segment VW in a 2:1 ratio.

$$\frac{1}{3}V + \frac{2}{3}W = \left(\frac{1}{3}(1) + \frac{2}{3}(9), \frac{1}{3}(6) + \frac{2}{3}(4) \right) = \left(6\frac{1}{3}, 4\frac{2}{3} \right)$$

Use the following information for questions 5-6. Line segment FW has endpoints at $F(-10, -10)$ and $W(6, -2)$.

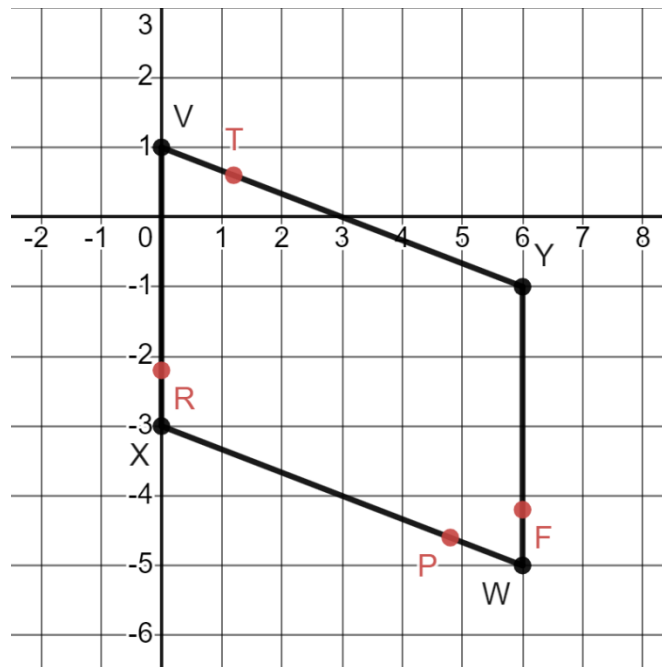
5. Write an expression for the point that partitions directed line segment FW in a ratio of 3 to 5.

$$\frac{5}{8}F + \frac{3}{8}W$$

6. Use the expression to find the point that partitions directed line segment FW in a 3:5 ratio.

$$\left(\frac{5}{8}(-10) + \frac{3}{8}(6), \frac{5}{8}(-10) + \frac{3}{8}(-2)\right) = (-4, -7)$$

Use the following information for questions 7-10. $VXWY$ has vertices at $V(0,1)$, $X(0,-3)$, $W(6,-5)$, and $Y(6,-1)$.



7. Find the point that partitions $\overline{WX}$ in a 1:4 ratio. Label it P .

$$\frac{4}{5}W + \frac{1}{5}X = \left(\frac{4}{5}(6) + \frac{1}{5}(0), \frac{4}{5}(-5) + \frac{1}{5}(-3)\right) = \left(4\frac{4}{5}, -4\frac{3}{5}\right) \text{ or } (4.8, -4.6)$$

8. Find the point that partitions $\overline{WY}$ in a 1:4 ratio. Label it F .

$$\frac{4}{5}W + \frac{1}{5}Y = \left(\frac{4}{5}(6) + \frac{1}{5}(6), \frac{4}{5}(-5) + \frac{1}{5}(-1)\right) = (6, -4.2)$$

9. Find the point that partitions $\overline{XV}$ in a 1:4 ratio. Label it R .

$$\frac{4}{5}X + \frac{1}{5}V = \left(\frac{4}{5}(0) + \frac{1}{5}(0), \frac{4}{5}(-3) + \frac{1}{5}(1)\right) = (0, -2.2)$$

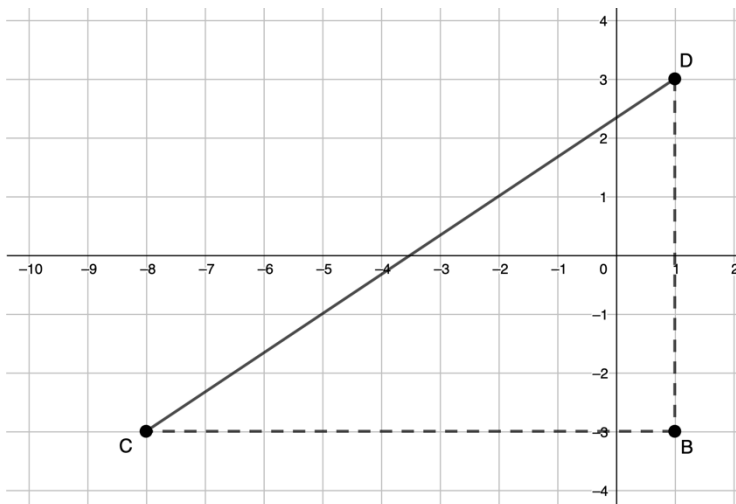
10. Find the point that partitions $\overline{VY}$ in a 1:4 ratio. Label it T .

$$\frac{4}{5}V + \frac{1}{5}Y = \left(\frac{4}{5}(0) + \frac{1}{5}(6), \frac{4}{5}(1) + \frac{1}{5}(-1)\right) = (1.2, 0.6)$$

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Lesson 6 Practice

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1. Find the length of segment CD using the Pythagorean Theorem. If necessary, round to the nearest thousandth.



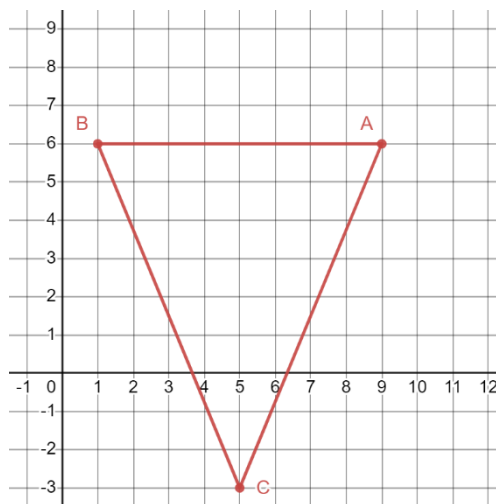
$$\begin{aligned} &C(-8, -3), D(1, 3) \\ &\sqrt{(-8 - 1)^2 + (-3 - 3)^2} \\ &= \sqrt{(-9)^2 + (-6)^2} \\ &= \sqrt{81 + 36} = \sqrt{117} \\ &= 3\sqrt{13} \approx 10.817 \end{aligned}$$

$$\overline{CD} = 10.817 \text{ units}$$

Use the following information for questions 2-6.

$\triangle ABC$ has vertices at $A(9,6)$, $B(1,6)$, and $C(5,-3)$.

2. Graph $\triangle ABC$ on the coordinate plane.



Find the length of each segment. If necessary, round to the nearest tenth.

	Line Segment	Length of Line Segment
3.	$\overline{AB}$	$9 - 1 = 8 \text{ units}$
4.	$\overline{BC}$	$\sqrt{(5 - 1)^2 + (-3 - 6)^2} = \sqrt{97} \approx 9.8 \text{ units}$
5.	$\overline{AC}$	$\sqrt{(5 - 9)^2 + (-3 - 6)^2} = \sqrt{97} \approx 9.8 \text{ units}$

6. Complete the statement.

$\triangle ABC$ is classified as a(n)

- ☐ equilateral
- ☒ isosceles
- ☐ scalene

triangle because $\triangle ABC$ has

- ☐ no
- ☒ two
- ☐ three

equal side lengths.

Use the following information for questions 7-10.

$\triangle TUV$ has vertices at $T(1, -8)$, $U(8, -1)$, $V(4, -1)$.

Find the length of each segment. If necessary, round to the nearest tenth.

	Line Segment	Length of Line Segment
7.	$\overline{TU}$	$\sqrt{(8-1)^2 + (-1-(-8))^2} = \sqrt{98} \approx 9.9$ units
8.	$\overline{UV}$	$8-4 = 4$ units
9.	$\overline{TV}$	$\sqrt{(4-1)^2 + (-1-(-8))^2} = \sqrt{58} \approx 7.6$ units

10. Complete the statement.

$\triangle TUV$ is classified as a(n)

- ☐ equilateral
- ☐ isosceles
- ☒ scalene

triangle because $\triangle TUV$ has

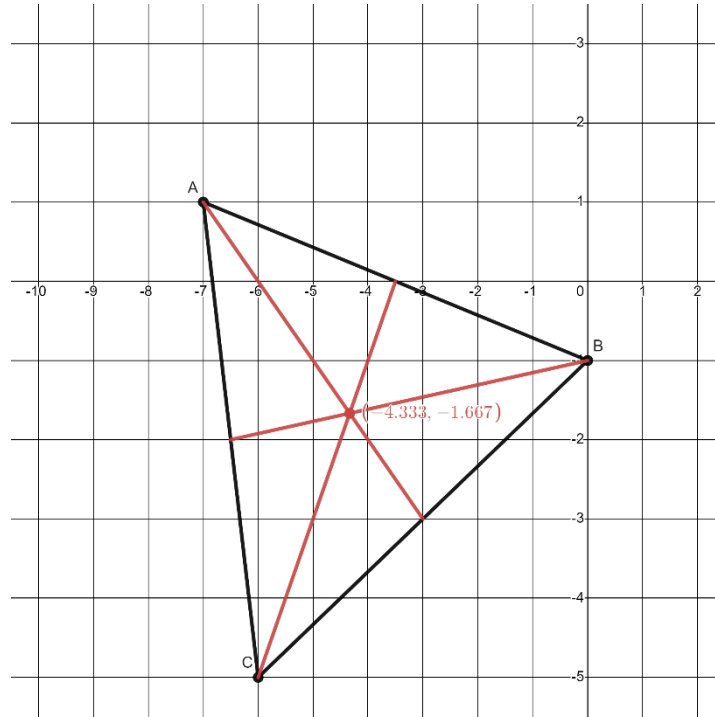
- ☒ no
- ☐ two
- ☐ three

equal side lengths.

Geometry
Unit 13
Lesson 7 Practice

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Use the following information for questions 1-4. $\triangle ABC$ has vertices at $A(-7,1)$, $B(0,-1)$, and $C(-6,-5)$.



1. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.

$$\frac{-7+0+(-6)}{3} = -4.333$$

2. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.

$$\frac{1+(-1)+(-5)}{3} = -1.667$$

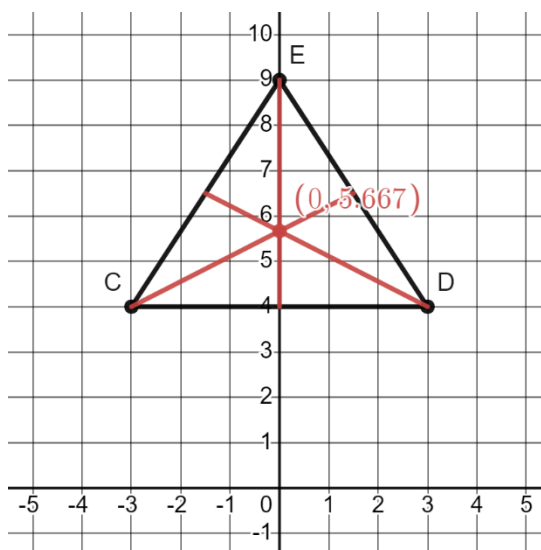
3. What are the coordinates of the centroid?

$$(-4.333, -1.667)$$

4. Verify the coordinates of the centroid by drawing the median of each side.

See graph.

$\triangle CDE$ has vertices at $C(-3,4)$, $D(3,4)$, and $E(0,9)$.



5. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.

$$\frac{-3+3+0}{3} = 0$$

6. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.

$$\frac{4+4+9}{3} = 5.667$$

7. What are the coordinates of the centroid of $\triangle CDE$?

$(0, 5.667)$

8. Verify the coordinates of the centroid by drawing the median of each side.

See graph.

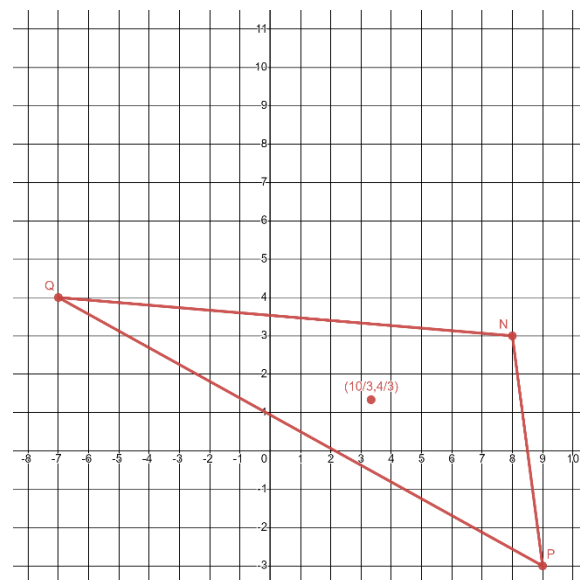
$\triangle NPQ$ has vertices at $N(8,3)$, $P(9,-3)$, and a centroid at $(\frac{10}{3}, \frac{4}{3})$.

9. Graph $\triangle NPQ$ on the coordinate plane.

See graph.

10. What are the coordinates of vertex Q ?

$$\begin{aligned} \frac{8+9+x}{3} &= \frac{10}{3} \\ 17+x &= 10 \\ x &= -7 \\ \frac{3+(-3)+y}{3} &= \frac{4}{3} \\ y &= 4 \\ (-7, 4) \end{aligned}$$

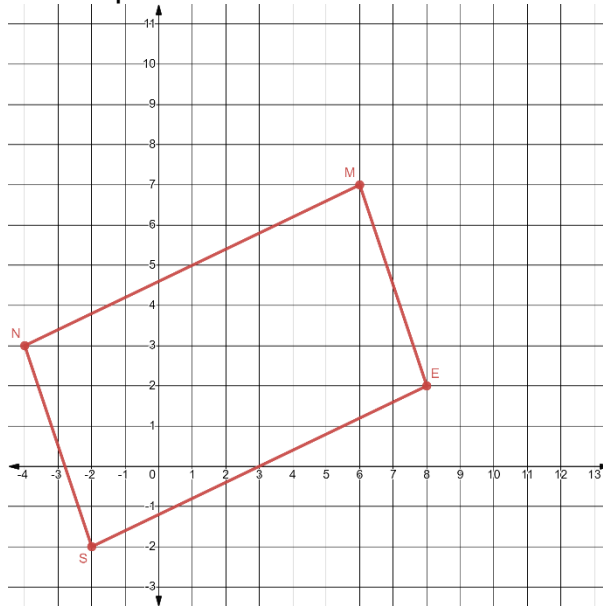


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Lesson 8 Practice

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$MNSE$ has vertices at $M(6,7)$, $N(-4,3)$, $S(-2,-2)$, and $E(8,2)$.

1. Graph $MNSE$ on the coordinate plane.



2. Complete the table.

Line Segment	Length	Slope	Length of Diagonal Line Segment
$\overline{MN}$	$\sqrt{4^2 + 10^2} = \sqrt{116}$ $= 2\sqrt{29}$	$\frac{4}{10} = \frac{2}{5}$	
$\overline{NS}$	$\sqrt{5^2 + 2^2} = \sqrt{29}$	$-\frac{5}{2}$	
$\overline{SE}$	$\sqrt{10^2 + 4^2} = \sqrt{116}$ $= 2\sqrt{29}$	$\frac{4}{10} = \frac{2}{5}$	
$\overline{ME}$	$\sqrt{2^2 + 5^2} = \sqrt{29}$	$-\frac{5}{2}$	
$\overline{MS}$			$\sqrt{8^2 + 9^2} = \sqrt{145}$
$\overline{NE}$			$\sqrt{12^2 + 1^2} = \sqrt{145}$

3. Circle all classifications that apply to $MNSE$.

Isosceles Trapezoid

Kite

Parallelogram

Rectangle

Rhombus

Square

Trapezoid

Complete the statements for questions 4 and 5.

Since $MNSE$ 4. ✗ has
o does not have congruent diagonals, $MNSE$ is classified as a(n)

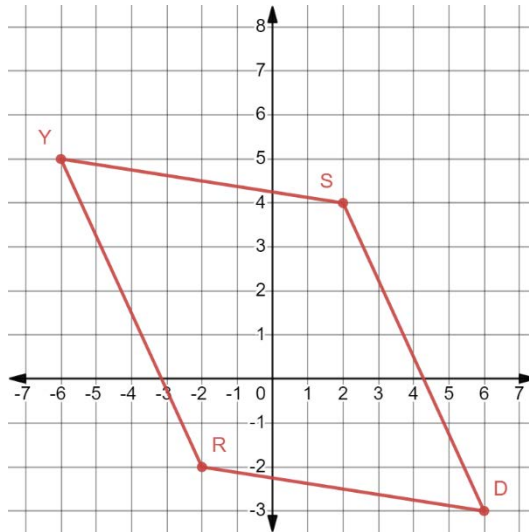
5. **rectangle**.

$YRDS$ has vertices at $Y(-6,5)$, $R(-2,-2)$, $D(6,-3)$, and $S(2,4)$.

$$YR = SD = YS = RD = \sqrt{65}$$

$$YD = 4\sqrt{13}$$

$$RS = 2\sqrt{13}$$



6. $YRDS$ is classified as a **rhombus** because... **all sides have equal length, but diagonals do not.**

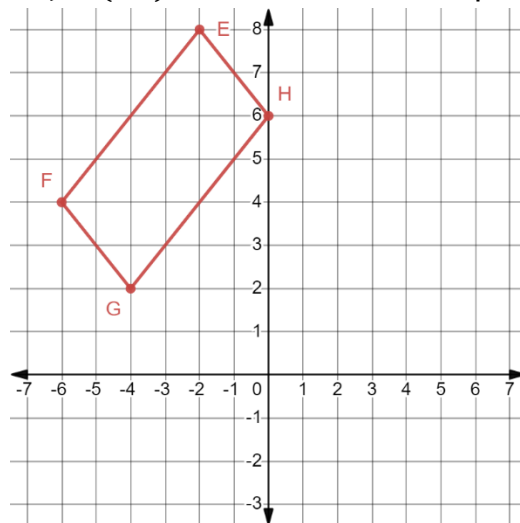
Jeferson graphed $E(-2,8)$, $F(-6,4)$, $G(-4,2)$, $H(0,6)$ on the coordinate plane and classified the quadrilateral as a square.

$$\overline{EH} \parallel \overline{FG}$$

$$\overline{FE} \parallel \overline{GH}$$

$$\overline{FG} \perp \overline{GH}$$

$$\overline{EH} \perp \overline{EF}$$



7. Plot the coordinates Jeferson graphed for his quadrilateral.
See graph.
8. The quadrilateral Jeferson graphed is a **rectangle** because ... **opposite sides are parallel and all angles are right angles.**
9. Do you agree or disagree with Jeferson? If you disagree, explain a reason Jeferson could have used to classify the quadrilateral as a square.
No, he could have said all angles are 90° , but all sides are not congruent.

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Unit 13
Lesson 9 Practice

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Najira is using a computer software to design the layout of her bathroom remodel. The base of the bathroom creates $DEFG$ with coordinates $D(7, -2)$, $E(1, -6)$, $F(3, -9)$, and $G(9, -5)$.

1. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DE}$	$DE = \sqrt{(1 - 7)^2 + (-6 + 2)^2} = \sqrt{(-6)^2 + (-4)^2} = 2\sqrt{13}$
$\overline{EF}$	$EF = \sqrt{(3 - 1)^2 + (-9 + 6)^2} = \sqrt{13}$
$\overline{FG}$	$FG = \sqrt{(9 - 3)^2 + (-5 + 9)^2} = 2\sqrt{13}$
$\overline{DG}$	$DG = \sqrt{(9 - 7)^2 + (-5 + 2)^2} = \sqrt{13}$

2. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DF}$	$DF = \sqrt{(3 - 7)^2 + (-9 + 2)^2} = \sqrt{65}$
$\overline{EG}$	$EG = \sqrt{(9 - 1)^2 + (-5 + 6)^2} = \sqrt{65}$

3. Finish the statement.

$DEFG$ is classified as a **rectangle** because ... **both pairs of opposite sides are congruent, diagonals are congruent.**

4. If 1 unit on the coordinate plane equals 3 feet (ft), what are the dimensions of the base of the bathroom? Round to the nearest thousandth.

Length: $2\sqrt{13}(3) = 6\sqrt{13} \approx 21.633$ ft.

Width: $\sqrt{13}(3) = 3\sqrt{13} \approx 10.817$ ft.

5. $PRNT$ has vertices at $P(3, -8)$, $R(2, 0)$, $N(-5, 4)$, and $T(-4, -4)$.

$PRNT$ is classified as a(n) **rhombus** because... **all sides are equal, diagonals are not equal.**

$$PR = RN = NT = TP = \sqrt{65}$$

$$PN = \sqrt{(-8)^2 + 12^2} = 4\sqrt{13}$$

$$RT = \sqrt{(-6)^2 + (-4)^2} = 2\sqrt{13}$$

Ginny is designing a template for her new 3D printer. She created two quadrilaterals using her software.

$MSFC$	$TDPR$
$M(0, -7), S(4, -3), F(6, 5), C(2, 1)$	$T(5, 7), D(7, 4), P(0, 4), R(2, 7)$

6. Which quadrilateral can be classified as a trapezoid? Show your work.

$TDPR$

$$TD = \sqrt{2^2 + (-3)^2} = \sqrt{13}$$

$$DP = \sqrt{(-7)^2 + 0^2} = 7$$

$$PR = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$RT = \sqrt{(-3)^2 + 0^2} = 3$$

Slopes:

$$\overline{TD}: m = -\frac{3}{2}$$

$$\overline{DP}: m = 0$$

$$\overline{PR}: m = \frac{3}{2}$$

$$\overline{RT}: m = 0$$

7. List the properties of a trapezoid that you used.

One pair of opposite sides are parallel (same slope).

8. If 1 unit of the coordinate plane equals 4 inches (in.), what are the measurements of the sides of the trapezoid?

$$TD = 4\sqrt{13} \approx 14.422 \text{ in.}$$

$$DP = 28 \text{ in.}$$

$$PR = 4\sqrt{13} \approx 14.422 \text{ in.}$$

$$RT = 12 \text{ in.}$$

Fintan was graphing square $RDCP$, but mixed up his vertices. He knows that point R has coordinates $(4, -5)$ and point P has coordinates $(2, -2)$. His two options for points C and D are shown.

Point C	Point D
$(-1, -4)$	$(-1, -7)$
$(-1, -3)$	$(1, -7)$

9. Which coordinates are correct for points C and D ? Justify your answer or show your work.

$$RP = \sqrt{13}, \text{ slope} = -\frac{3}{2}$$

$$RD = \sqrt{13}, \text{ slope} = \frac{2}{3}$$

$$PC = \sqrt{13}, \text{ slope} = \frac{2}{3}$$

$$C(-1, -4), D(1, -7)$$

10. Which properties of a square did you use to determine the correct coordinates for points C and D ?

Length of all 4 sides are congruent and slope of adjacent sides opposite reciprocal (sides perpendicular).

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Complete the table for each image in questions 1 and 2.

Question 1	Work		Answer
			Properties
	Slope	$\overline{MN} \text{ \& } \overline{RP}$: $m = \frac{6}{2} = 3$ $\overline{MR} \text{ \& } \overline{NP}$: $m = \frac{5}{7}$	Opposite sides are parallel and congruent.
	Side Lengths	$MN = \sqrt{6^2 + 2^2} = 2\sqrt{10}$ $RP = \sqrt{2^2 + 6^2} = 2\sqrt{10}$ $MR = \sqrt{5^2 + 7^2} = \sqrt{74}$ $NP = \sqrt{7^2 + 5^2} = \sqrt{74}$	
	Diagonal Lengths	$NR = \sqrt{9^2 + 11^2} = \sqrt{202}$ $MP = \sqrt{1^2 + 5^2} = \sqrt{26}$	Quadrilateral Parallelogram
Question 2	Work		Answer
			Properties
	Slope	$\overline{XM} \text{ \& } \overline{BY}$: $m = \frac{4}{6} = \frac{2}{3}$ $\overline{XB} \text{ \& } \overline{MY}$: $m = -\frac{6}{4} = -\frac{3}{2}$	Opposite sides are parallel and congruent. Consecutive sides are perpendicular (90°). Diagonals are congruent. All sides are equal.
	Side Lengths	$\overline{XM}, \overline{MY}, \overline{YB}, \overline{BX}$: $= \sqrt{4^2 + 6^2} = \sqrt{52}$ $= 2\sqrt{13}$	
	Diagonal Lengths	$MB = \sqrt{2^2 + 10^2} = 2\sqrt{26}$ $XY = \sqrt{10^2 + 2^2} = 2\sqrt{26}$	Quadrilateral Square

For questions 3 and 4, given the coordinates, classify the quadrilaterals based on their properties.

Question 3	Work		Answer
			Properties
$CMHG$ has vertices at $C(9, -1)$, $M(2, 5)$, $H(-7, 7)$, and $G(0, 1)$. $CMHG$ is classified as a(n) rhombus because... all sides are equal, consecutive sides are not perpendicular.	Slope	$\overline{CM} \text{ \& } \overline{HG}$: $m = -\frac{6}{7}$ $\overline{MH} \text{ \& } \overline{CG}$: $m = -\frac{2}{9}$	Opposite sides are parallel and congruent. All sides are congruent. Diagonals are not congruent.
	Side Lengths	$CM = \sqrt{(-7)^2 + 6^2} = \sqrt{85}$ $MH = \sqrt{(-9)^2 + 2^2} = \sqrt{85}$ $HG = \sqrt{(-7)^2 + 6^2} = \sqrt{85}$ $GC = \sqrt{9^2 + (-2)^2} = \sqrt{85}$	
	Diagonal Lengths	$CH = \sqrt{(-16)^2 + 8^2} = 8\sqrt{5}$ $MG = \sqrt{2^2 + 4^2} = 2\sqrt{5}$	Quadrilateral Rhombus

Question 4	Work		Answer
			Properties
$TCBY$ has vertices at $T(-3, -7)$, $C(5, -3)$, $B(3, 1)$, and $Y(-5, -3)$. $TCBY$ is classified as a(n) rectangle because... opposite sides are parallel and congruent and all angles are right angles.	Slope	$\overline{TC} \text{ \& } \overline{BY}$: $m = \frac{1}{2}$ $\overline{CB} \text{ \& } \overline{YT}$: $m = -2$	Opposite sides are parallel and congruent. Consecutive sides are perpendicular. Diagonals are congruent.
	Side Lengths	$TC = \sqrt{8^2 + 4^2} = 4\sqrt{5}$ $CB = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$ $BY = \sqrt{(-8)^2 + (-4)^2} = 4\sqrt{5}$ $YT = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$	
	Diagonal Lengths	$TB = \sqrt{6^2 + 8^2} = 10$ $CY = \sqrt{10^2 + 0^2} = 10$	Quadrilateral Rectangle

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Compare and contrast each image pair. Record your work and show your justifications in the table.

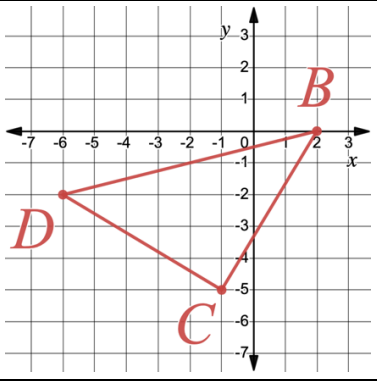
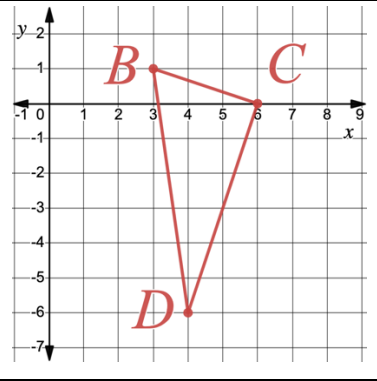
Image 1	Image 2
	
Coordinates: $B(2,0)$, $C(-1,-5)$, $D(-6,-2)$	Coordinates: $B(3,1)$, $C(6,0)$, $D(4,-6)$
Classification: Isosceles Right Triangle	Classification: Right Triangle
Property: $DC = BC$, $\overline{DC} \perp \overline{BC}$	Property: $\overline{DC} \perp \overline{BC}$
Similarity/Differences with Image 2: Both triangles have a right angle Two congruent sides	Similarity/Differences with Image 1: Both triangles have a right angle No congruent sides

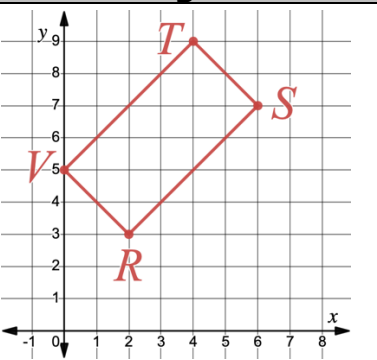
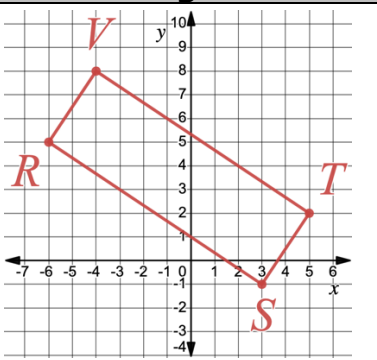
Image 3	Image 4
	
Coordinates: $R(2,3)$, $S(6,7)$, $T(4,9)$, $V(0,5)$	Coordinates: $R(-6,5)$, $S(3,-1)$, $T(5,2)$, $V(-4,8)$
Classification: Rectangle	Classification: Rectangle
Property: $\overline{VR} \perp \overline{RS}$, $\overline{VT} \perp \overline{TS}$, $\overline{VR} \cong \overline{TS}$, $\overline{VT} \cong \overline{RS}$	Property: $\overline{VR} \perp \overline{RS}$, $\overline{VT} \perp \overline{TS}$, $\overline{VR} \cong \overline{TS}$, $\overline{VT} \cong \overline{RS}$
Similarity/Differences with Image 2: Same shape and properties. The sides are not the same length.	Similarity/Differences with Image 1: Same shape and properties. The sides are not the same length.

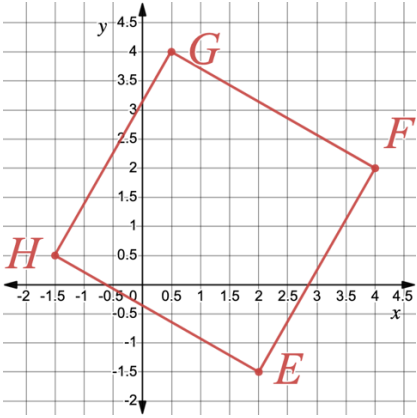
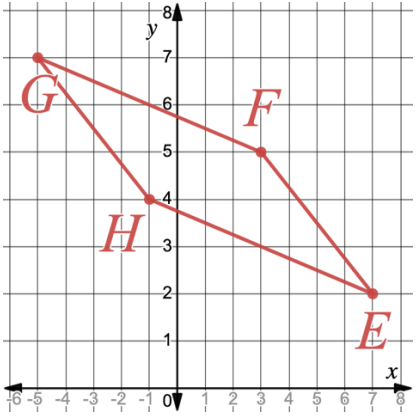
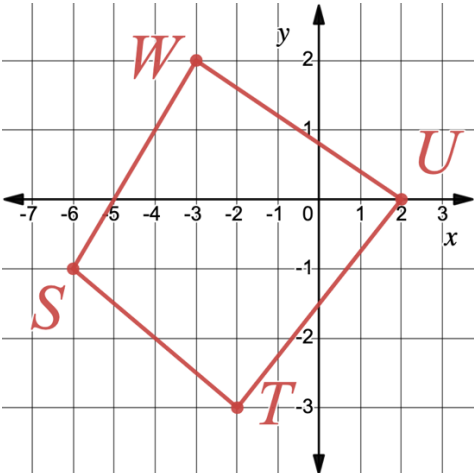
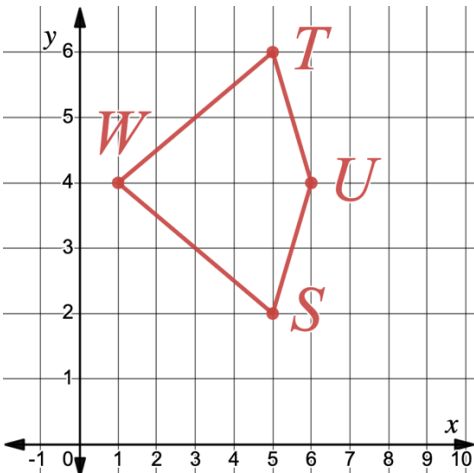
Image 5	Image 6
	
Coordinates: $E(2, -1.5), F(4, 2), G(0.5, 4), H(-1.5, 0.5)$	Coordinates: $E(7, 2), F(3, 5), G(-5, 7), H(-1, 4)$
Classification: Square	Classification: Parallelogram
Property: $\overline{GH} \perp \overline{GF}, \overline{HE} \perp \overline{EF}, \overline{GH} \cong \overline{EF}, \overline{FG} \cong \overline{EH}$	Property: $\overline{GH} \cong \overline{EF}, \overline{FG} \cong \overline{EH}$
Similarity/Differences with Image 2: All sides are congruent Opposite sides are parallel	Similarity/Differences with Image 1: All sides are not congruent Opposite sides are parallel

Image 7	Image 8
	
Coordinates: $S(-6, -1), T(-2, -3), U(2, 0), W(-3, 2)$	Coordinates: $S(5, 2), T(5, 6), U(6, 4), W(1, 4)$
Classification: Quadrilateral	Classification: Kite
Property: no sides are congruent or parallel	Property: $\overline{TU} \cong \overline{SU}, \overline{WS} \cong \overline{WT}$
Similarity/Differences with Image 2: There are 4 sides No sides are congruent	Similarity/Differences with Image 1: There are 4 sides Adjacent sides are congruent

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Right triangle MTH has vertices at $M(3, -8), T(8, 4), H(8, -8)$. Right triangle DPT has vertices at $D(-3, -4), P(-5, -6), T(-3, -6)$.

1. Determine if $\triangle MTH$ and $\triangle DPT$ are congruent, similar, or neither. **Neither**

$$MT = \sqrt{(8-3)^2 + (4+8)^2} = 13$$

$$DP = \sqrt{(-5+3)^2 + (-6+4)^2} = \sqrt{8} = 2\sqrt{2}$$

$$TH = \sqrt{(8-8)^2 + (4+8)^2} = 12$$

$$PT = \sqrt{(-3+5)^2 + 0^2} = \sqrt{4} = 2$$

$$HM = \sqrt{(8-3)^2 + 0^2} = 5$$

$$TD = \sqrt{(-6+4)^2 + (0)^2} = \sqrt{4} = 2$$

2. What definition, theorem, or property justifies your conclusions?

Does not satisfy SSS congruence or similarity.

3. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.

Using triangle MTH as the pre-image, shift 11 units left and 2 units up. This will align the right angles, and you can see the triangles are neither congruent or similar.

Determine if the given polygons are congruent, similar, or neither.

4. Trapezoid $TRPS$: $T(1,1), R(4,5), P(7,5), S(10,1)$
Trapezoid $RTLF$: $R(3,3), T(12,15), L(21,15), F(30,3)$

- | | | |
|--|----------------------------|------------------------------|
| ☐ Congruent | $TR: \sqrt{3^2 + 4^2} = 5$ | $RT: \sqrt{9^2 + 12^2} = 15$ |
| ☒ Similar | $RP: \sqrt{0^2 + 3^2} = 3$ | $TL: \sqrt{9^2} = 9$ |
| ☐ Neither | $PS: \sqrt{3^2 + 4^2} = 5$ | $LF: \sqrt{9^2 + 12^2} = 15$ |
| | $TS: \sqrt{9^2 + 0^2} = 9$ | $RF: \sqrt{27^2} = 27$ |

5. Kite $GHKM$: $G(0,0), H(1,-4), K(3,-3), M(4,-1)$.
Kite $RSTV$: $R(0,0), S(2,-8), T(6,-6), V(8,-2)$.

- | | | |
|--|-----------------------------------|---|
| ☐ Congruent | $GH: \sqrt{1^2 + 16} = \sqrt{17}$ | $RS: \sqrt{2^2 + 8^2} = \sqrt{68} = 2\sqrt{17}$ |
| ☒ Similar | $HK: \sqrt{4 + 1} = \sqrt{5}$ | $ST: \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$ |
| ☐ Neither | | |

2:1 ratio

6. Rhombus $DEFG$: $D(6,4), E(-1,0), F(-2,-8), G(5,-4)$
Rhombus $BKMH$: $B(3,2), K(-0.5,0), M(-1,-4), H(2.5,-2)$

- | | |
|--|--|
| ☐ Congruent | Corresponding angles are congruent. Since sides are congruent, ratios of corresponding sides are congruent. |
| ☒ Similar | |
| ☐ Neither | |

Lane and Tonymy are determining if parallelograms $HMTY$ and $NRPX$ are congruent, similar, or neither. Parallelogram $HMTY$ has vertices at $H(5,6)$, $M(-3,3)$, $T(-4,-1)$, and $Y(4,2)$. Parallelogram $NRPX$ has vertices at $N(10,12)$, $R(-6,6)$, $P(-8,-2)$, and $X(8,4)$. Their work is shown.

Lane	Tonymy
Parallelograms $HMTY$ and $NRPX$ are similar because $\frac{MH}{RN} = \frac{XR}{PN}$	Parallelograms $HMTY$ and $NRPX$ are not similar because $\frac{MH}{RN} = \frac{XN}{PR}$

7. Who do you agree with?

Neither, Lane used diagonals instead of sides, and Tonymy did not setup the proportion correctly.

8. Explain the mistake that was made and how to correct it.

$$\frac{HM}{RN} = \frac{\sqrt{8^2 + 3^2}}{\sqrt{16^2 + 6^2}} = \frac{\sqrt{73}}{\sqrt{292}}$$

$$\frac{YH}{XN} = \frac{\sqrt{1^2 + 4^2}}{\sqrt{2^2 + 8^2}} = \frac{\sqrt{17}}{\sqrt{68}}$$

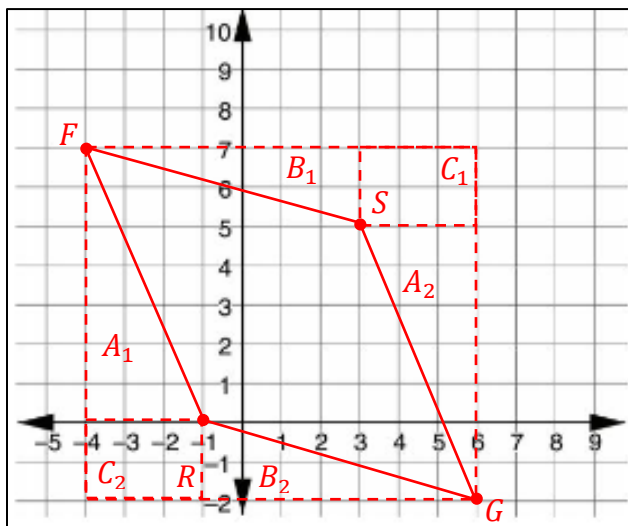
$$\frac{\sqrt{73}}{\sqrt{292}} = \frac{\sqrt{17}}{\sqrt{68}}$$

Explanations may vary but should include that the parallelogram are similar due to having equivalent ratios for the corresponding sides.

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Parallelogram $FRGS$ has vertices at $F(-4,7)$, $R(-1,0)$, $G(6,-2)$, and $S(3,5)$.



- Graph $FRGS$.
- Complete the table.

Line Segment	Length
$\overline{FR}$	$FR = \sqrt{7^2 + 3^2} = \sqrt{58}$
$\overline{RG}$	$RG = \sqrt{2^2 + 7^2} = \sqrt{53}$
$\overline{GS}$	$GS = \sqrt{3^2 + 7^2} = \sqrt{58}$
$\overline{FS}$	$FS = \sqrt{7^2 + 2^2} = \sqrt{53}$

- What is the perimeter of $FRGS$?
 $P = \sqrt{58} + \sqrt{53} + \sqrt{58} + \sqrt{53} = 2\sqrt{58} + 2\sqrt{53} \approx 29.79$

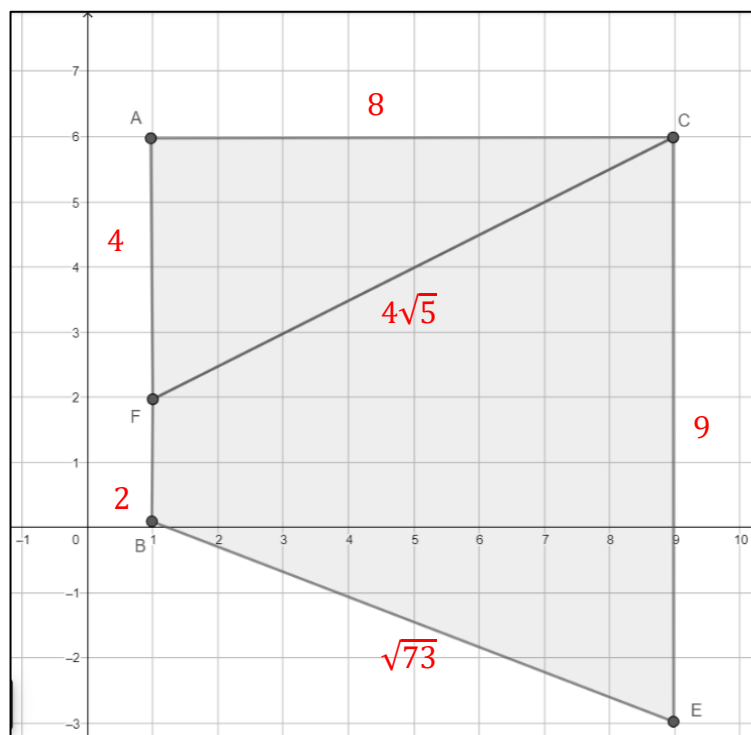
- What is the area of $FRGS$?

$$A_{TOTAL} = (9)(10) = 90 \quad \therefore A_{FRGS} = 90 - 2(10.5 + 7 + 6) = 43 \text{ units}^2$$

$$\text{Area of Regions:}$$

$$A_1 = A_2 = \frac{1}{2}(7)(3) = 10.5 \quad \left| \quad B_1 = B_2 = \frac{1}{2}(7)(2) = 7 \quad \right| \quad C_1 = C_2 = (3)(2) = 6$$

The composite figure shown is created by $A(1,6)$, $B(1,0)$, $C(9,6)$, $F(1,2)$, and $E(9,-3)$.



5. Complete the table. Round your answer to the nearest thousandth.

Figure	Perimeter	Area
$\triangle ACF$	$FC = \sqrt{4^2 + 8^2} = \sqrt{80} = 4\sqrt{5}$ $P = 4 + 8 + 4\sqrt{5} \approx 20.944$	$A = \frac{1}{2}(4)(8) = 16 \text{ units}^2$
$BECF$	$BE = \sqrt{3^2 + 8^2} = \sqrt{73}$ $P = 2 + 4\sqrt{5} + 9 + \sqrt{73} \approx 28.488$	$A = \frac{1}{2}(8)(2 + 9)$ $A = 44 \text{ units}^2$

6. What is the perimeter of the composite figure?

$$P = 8 + 4 + 2 + 9 + \sqrt{73} = 23 + \sqrt{73} \approx 31.544 \text{ units}$$

7. What is the area of the composite figure?

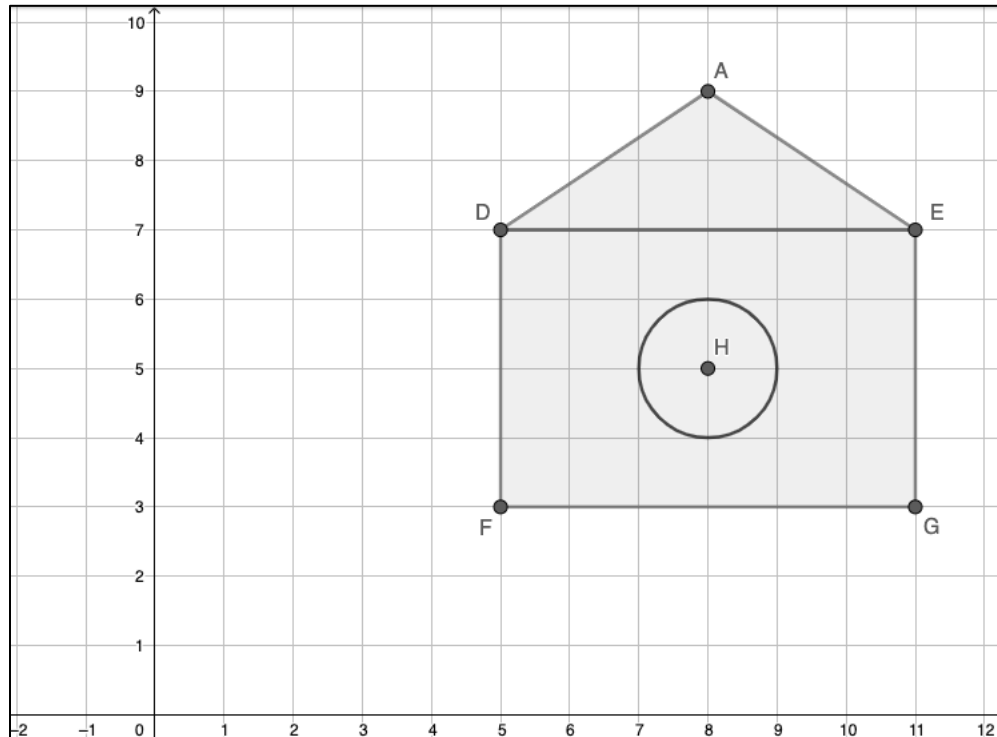
$$\text{Total Area:}$$

$$16 + 44 = 60 \text{ units}^2$$

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Rory is designing the front of a birdhouse. The computer software generated coordinate points where each unit equals an inch. The roof has vertices at vertices at $A(8,9)$, $D(5,7)$, and $E(11,7)$. The front entrance has a circle with a radius of 1 unit. The front wall has vertices at $D(5,7)$, $E(11,7)$, $F(5,3)$, and $G(11,3)$.



- Classify the type of triangle that is on the roof?

$$DE = 6$$

$$DA = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$EA = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$\triangle DAE$ is an isosceles triangle

- What is the total area of the birdhouse?

$$\text{Area of Triangle} = \frac{1}{2}(6)(2) = 6 \text{ in}^2$$

$$\text{Area of Square} = (6)(4) = 24 \text{ in}^2$$

$$\text{Area of Circle} = \pi(1)^2 = 3.14 \text{ in}^2$$

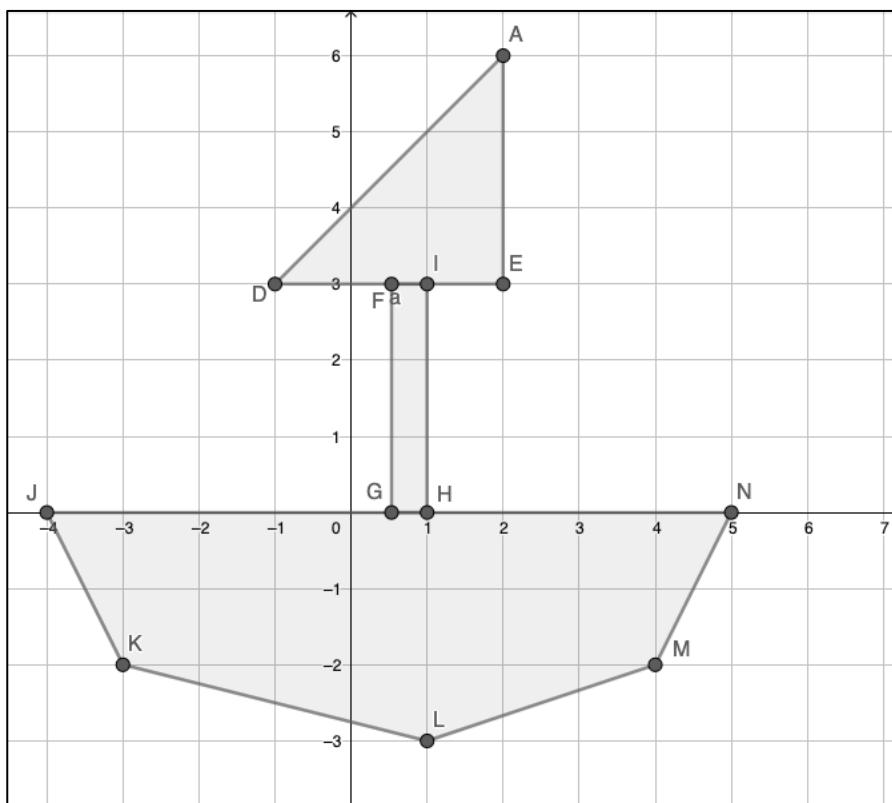
$$\text{Total Area} = 24 + 6 - 3.14 = 26.86 \text{ in}^2$$

- Cartwright wants to use weather-stripping around the entrance. It costs \$0.15 per inch. How much weather-stripping is going to be needed around the entrance to the bird house?

$$\text{Circumference of Circle} = 2\pi(1) = 6.28 \text{ in}^2$$

$$\text{Cost of weather stripping} = 6.28 \cdot 0.15 = \$0.94$$

Kima and Whuan are building a sailboat by hand. The layout of the sailboat is sketched on the blueprint shown.



4. If one square on the grid represents two feet, what is the distance around the sailboat on the blueprint?

$$JN = 9$$

$$JK = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$KL = \sqrt{1^2 + 4^2} = \sqrt{17}$$

$$LM = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$MN = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$P = 9 + \sqrt{5} + \sqrt{7} + \sqrt{10} + \sqrt{5}$$

$$P \approx 20.76 \cdot 2 \approx 41.52 \text{ ft}$$

5. If one square on the grid represents three square feet, what is the total area of the sailboat on the blueprint?

$$\text{Area of } DAE = \frac{1}{2}(3)(3) = 4.5 \text{ units}^2$$

$$\text{Area of } JKMN = \frac{1}{2}(2)(9 + 7) = 16 \text{ units}^2$$

$$\text{Area of } FGHI = (0.5)(3) = 1.5 \text{ units}^2$$

$$\text{Area of } KLM = \frac{1}{2}(7)(1) = 3.5 \text{ units}^2$$

$$\text{Total Area} = 4.5 + 1.5 + 16 + 3.5 = 25.5 \text{ units}^2 = 25.5 \cdot 3 = 76.5 \text{ ft}^2$$